

# Supplemental Materials

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## A Appendix

We reiterate the classic Central Limit Theorem (CLT) below, which is central to our proofs below.

**THEOREM A.1.** *Lindeberge-Lévy CLT [1, Thm. 27.1].*

Let  $\{X_1, \dots, X_n, \dots\}$  be an independent sequence of random variables with the same distribution, whose mean is  $\mu$  and has a finite variance  $\sigma^2$ . If  $S_n = \sum_{i=1}^n X_i$ , then  $\frac{S_n - n\mu}{\sigma\sqrt{n}}$  converges to a standard normal distribution  $N(0, 1)$ .

### A.1 Correctness of combined estimator $\tilde{A}'$ and confidence interval $\varepsilon'$ in Section 3

Let  $\tilde{A}_1, \dots, \tilde{A}_k$  be the independent estimators derived from the  $k$  approximate subqueries  $\tilde{Q}_1, \dots, \tilde{Q}_k$ . Recall that  $\tilde{A}'$  is the sum of these estimators. By the linearity of expectation, we have

$$E[\tilde{A}'] = E\left[\sum_{i=1}^k \tilde{A}_i\right] = \sum_{i=1}^k E[\tilde{A}_i] = \sum_{i=1}^k A_i = A$$

For the overall half-width of confidence interval  $\varepsilon$ 's, we note that  $\tilde{A}_k$  are  $k$  independent random variables that all converge to (different) normal distributions  $\mathcal{N}(\tilde{A}_i, \frac{\sigma_i^2}{n_i})$  (Theorem A.1), where each  $\sigma_i$  is an unknown constant that can be estimated using Equation 4 in Section 2. Then the final estimator  $\tilde{A}$  follows a mixed normal distribution  $\mathcal{N}(A, \sum_{i=1}^k \frac{\sigma_i^2}{n_i})$ . Hence, the CI  $\varepsilon$  can be estimated by solving the following:

$$\Pr\{|\tilde{A}' - A| \leq \varepsilon'\} = 1 - \delta \Rightarrow \varepsilon = Z_\delta \sqrt{\sum_{i=1}^k \frac{\sigma_i^2}{n_k}} = Z_\delta \sqrt{\sum_{i=1}^k \frac{\varepsilon_i^2 n_i / Z_\delta^2}{n_k}} = \sqrt{\sum_{i=1}^k \varepsilon_i^2}$$

### A.2 Proofs of Lemmas 3.1 and 3.2

We first prove lemma 3.2 below.

**LEMMA 3.2. (Modified Neyman allocation)** *Given a set of strata  $D = D_1 \uplus \dots \uplus D_k$  and the estimator variances  $\sigma_1^2, \dots, \sigma_k^2$  within each stratum, the minimum index-assisted sampling cost for approximate query  $\tilde{Q}'$  to reach a  $(\varepsilon, \delta)$  confidence bound is  $c = c_0 k + \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i \sqrt{h_i})^2$ , where the sample size for each stratum is  $n_i = \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i \sqrt{h_i}) \frac{\sigma_i}{\sqrt{h_i}}$ .*

**PROOF.** We want to minimize  $c = c_0 k + \sum_{i=1}^k n_i \cdot h_i$ , subject to  $\varepsilon = Z_\delta \sqrt{\sum_{i=1}^k \frac{\sigma_i^2}{n_i}}$ . To do so, we apply the Lagrange multiplier method as follows:

Let  $\lambda$  be the Lagrange multiplier. The target function becomes  $F = c_0 k + \sum_{i=1}^k n_i \cdot h_i - \lambda (\frac{\varepsilon^2}{Z_\delta^2} - \sum_{i=1}^k \frac{\sigma_i^2}{n_i})$ .

Then we take the partial derivatives for  $\lambda$  and each  $n_i$  as follows:

$$\frac{\partial F}{\partial \lambda} = -\frac{\varepsilon^2}{Z_\delta^2} + \sum_{i=1}^k \frac{\sigma_i^2}{n_i}$$

$$\frac{\partial F}{\partial n_i} = h_i + (-\lambda \frac{\varepsilon^2}{Z_\delta^2})$$

Now, solve for  $n_i$  and  $\lambda$  when all partial derivatives are zero, which is when the minimum happens:

$$\frac{\partial F}{\partial n_i} = 0 \Rightarrow n_i = \frac{\sqrt{\lambda} \sigma_i}{\sqrt{h_i}}$$

$$\frac{\partial F}{\partial \lambda} = 0 \Rightarrow \frac{\varepsilon^2}{Z_\delta^2} = \sum_{i=1}^k \frac{\sigma_i^2}{n_i}$$

$$\text{As a result, } n_i = \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i \sqrt{h_i}) \frac{\sigma_i}{\sqrt{h_i}}$$

$$c = c_0 k + \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i \sqrt{h_i})^2 \quad \square$$

**LEMMA 3.1. (Neyman allocation)** *Given a set of strata  $D = D_1 \uplus \dots \uplus D_k$  and the estimator variances  $\sigma_1^2, \dots, \sigma_k^2$  within each stratum, the minimum total sample size for approximate query  $\tilde{Q}'$  to reach a  $(\varepsilon, \delta)$  confidence bound is  $n' = \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i)^2$ , where the sample size for each stratum is  $n_i = \frac{Z_\delta^2}{\varepsilon^2} (\sum_{i=1}^k \sigma_i) \sigma_i$ .*

**PROOF.** This is true by letting  $h_i = 1$  for all  $i$  in Lemma 3.2.  $\square$

### A.3 Proof of Theorem 3.3

To prove Theorem 3.3, it suffices to prove the following Lemma A.2 based on our proof sketch following Theorem 3.3:

**LEMMA A.2.** *For any 2 disjoint stratification of the range  $D = D_1 \uplus D_2$  where  $D_1, D_2 \neq \emptyset$ , the sum of estimator standard deviation  $\sigma' = \sigma_1 + \sigma_2$  for index-assisted stratified sampling in  $\tilde{Q}'$  is no larger than  $\sigma$  – the estimator standard deviation using one single stratum  $D$ , for the same confidence interval  $\varepsilon$  at confidence level  $1 - \delta$ .*

**PROOF.** Our first step is to quantify the three variance terms  $\sigma^2, \sigma_1^2$  and  $\sigma_2^2$ . To do so, we denote the fraction of records in  $D_1$  be  $\alpha_1 = |\sigma_{x \in D_1} T| / |\sigma_{x \in D} T|$ , and thus the fraction of records in  $D_2$  is  $\alpha_2 = 1 - \alpha_1$ . Let the total number of records in the query range be  $N = |\sigma_{x \in D} T|$ . Next, we observe that  $\sigma_i^2 (\forall i \in \{1, 2\})$  is the variance of the individual unbiased estimators for each random sample,  $\tilde{A}(t)$  as defined in Equation 1, with<sup>1</sup>  $p_i = \frac{1}{\alpha_i N}$ . If we denote a sampled tuple as  $t$ , then it can be theoretically computed as

<sup>1</sup>For ease of illustration, we assume uniform sampling, but the proof with weighted sampling can be similarly derived using the generalized Jensen's inequality, which we omit for brevity.

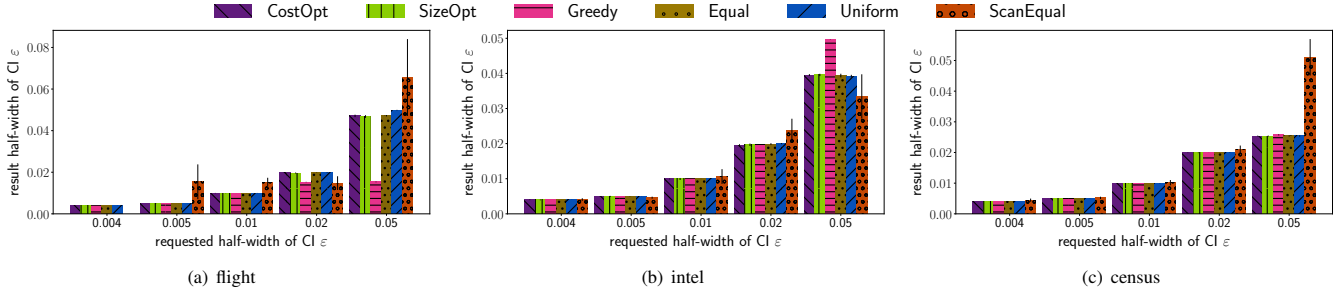


Figure 1: Reported confidence interval with varying requested confidence interval.

$$\begin{aligned}
 \sigma_i^2 &= E[\tilde{A}(t)^2] - E^2[\tilde{A}(t)] \\
 &= \sum_{t \in D_i} \frac{\tilde{A}_{\mathcal{P}_f, p_i}^2(t)}{\alpha_i N} - \left( \frac{\sum_{t \in D_i} \tilde{A}_{\mathcal{P}_f, p_i}(t)}{\alpha_i N} \right)^2 \\
 &= (\alpha_i N)^2 \left[ \sum_{t \in D_i} \frac{(e(t) \mathcal{P}_f(t))^2}{\alpha_i N} - \left( \sum_{t \in D_i} \frac{e(t) \mathcal{P}_f(t)}{\alpha_i N} \right)^2 \right]
 \end{aligned}$$

Similarly, the variance of each individual estimator is the following, where  $p(t) = \frac{1}{N}$ :

$$\sigma^2 = N^2 \left[ \sum_{t \in D} \frac{(e(t) \mathcal{P}_f(t))^2}{N} - \left( \sum_{t \in D} \frac{e(t) \mathcal{P}_f(t)}{N} \right)^2 \right]$$

Now we can use the Jensen's inequality to show the following, as square root is a concave function:

$$\begin{aligned}
 \sigma_1 + \sigma_2 &= \sum_{i \in \{1,2\}} \alpha_i N \sqrt{\sum_{t \in D_i} \frac{(e(t) \mathcal{P}_f(t))^2}{\alpha_i N} - \left( \sum_{t \in D_i} \frac{e(t) \mathcal{P}_f(t)}{\alpha_i N} \right)^2} \\
 &\leq N \sqrt{\sum_{i \in \{1,2\}} \alpha_i \cdot \left( \sum_{t \in D_i} \frac{(e(t) \mathcal{P}_f(t))^2}{\alpha_i N} - \left( \sum_{t \in D_i} \frac{e(t) \mathcal{P}_f(t)}{\alpha_i N} \right)^2 \right)} \\
 &= N \sqrt{\sum_{t \in D} \frac{(e(t) \mathcal{P}_f(t))^2}{N} - \sum_{i \in \{1,2\}} \alpha_i \cdot \left( \sum_{t \in D_i} \frac{e(t) \mathcal{P}_f(t)}{\alpha_i N} \right)^2}
 \end{aligned}$$

By applying Jensen's inequality again, observing that square is a convex function, we complete the proof:

$$\begin{aligned}
 \sigma_1 + \sigma_2 &\leq N \cdot \sqrt{\sum_{t \in D} \frac{(e(t) \mathcal{P}_f(t))^2}{N} - \left( \sum_{i \in \{1,2\}} \alpha_i \sum_{t \in D_i} \frac{e(t) \mathcal{P}_f(t)}{\alpha_i N} \right)^2} \\
 &= \sigma
 \end{aligned}$$

□

#### A.4 Computing unbiased estimator and confidence intervals with substrata

To deal with the combination with samples from stratum  $D$ , and samples from each new strata  $D' \in D$ , we update the mean and variance using the following formulas:

$$\mu' = \mu + \frac{\Delta k}{\Delta k + 1} (-\mu_D + \sum_{i=1}^{\Delta k} \mu_{D'_k})$$

$$\tilde{\sigma}'^2 = \tilde{\sigma}^2 + \left( \frac{1}{\Delta k} \right)^2 \sum_{i=1}^{\Delta k} \tilde{\sigma}_{D'_k}^2 - \left( \frac{\Delta k - 1}{\Delta k} \right)^2 \tilde{\sigma}_D^2$$

#### A.5 Cumulative sum statistics for computing stratum estimator variance

To deal with the problem of floating-point error, we update the mean and variance when each new tuple comes into accumulation function as aggregated values instead of using  $Sx$  and  $Sxx$ .

In the previous function,

$$Sx[j] = \sum_{t \in S_{i,j}} \frac{e(t)}{p(t)}$$

$$Sxx[j] = \sum_{t \in S_{i,j}} \left( \frac{e(t)}{p(t)} - \frac{Sx}{n_{2:,j}} \right)^2$$

where  $i$  is the  $i^{th}$  phase,  $j$  is the  $j^{th}$  stratum in the phase, and  $n_{i,j} = |S_{i,j}|$ , which is the sample size of the stratum  $j$  in phase  $i$ .

In the new function, we defined the aggregated values as follows:

$$\mu_x[j] = \frac{Sx[j]}{n_{2:,j}}$$

$$\tilde{\sigma}^2[j] = \frac{Sxx}{n_{2:,j}(n_{2:,j} - 1)}$$

When the new tuple comes, they will be updated using the following formulas:

$$\begin{aligned}
 \mu_x[j]' &= \frac{n_{2:,j}}{n_{2:,j} + 1} \mu_x[j] + \frac{e(t)/p(t)}{n_{2:,j} + 1} \\
 \tilde{\sigma}'^2[j] &= \frac{(e(t)/p(t) - \mu_x[j])^2}{(n_{2:,j} + 1)^2} + \frac{n_{2:,j} - 1}{n_{2:,j} + 1} \tilde{\sigma}^2[j]
 \end{aligned}$$

#### A.6 Reported confidence interval with varying requested confidence interval on real-world datasets

On real-world datasets, flight, intel and census, we verified that all methods except ScanEqual are returning confidence intervals below the requested confidence intervals (Figure 1). Note that the error bars for all methods in OptiAQP are quite small, indicating that for these high-variance queries, OptiAQP can produce quite stable estimations.

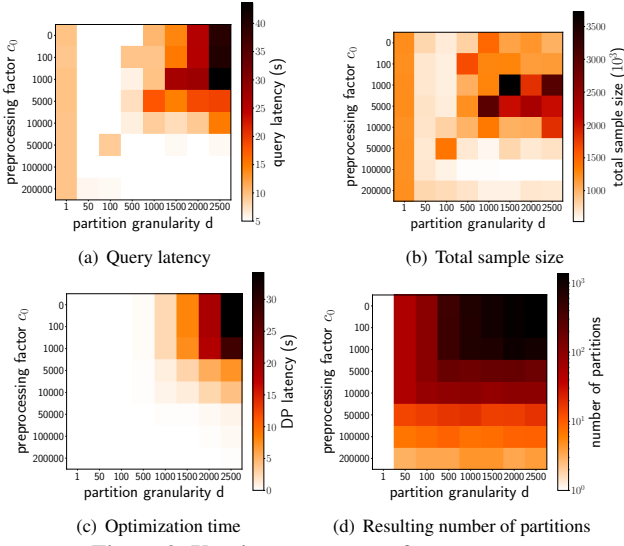


Figure 2: Varying parameters for CostOpt

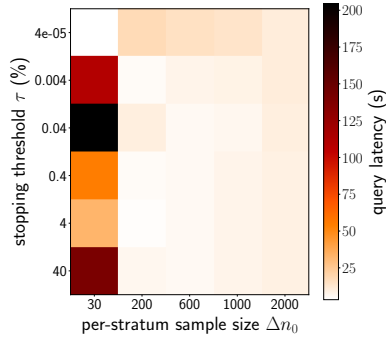


Figure 3: Varying parameters for Greedy

## A.7 Hyper-parameter selection

The selection of hyper-parameters is guided by the trade-offs between optimization overhead and benefit. As demonstrated in Figure 2 and Figure 3, our methods remain robust performance across a large amount of hyper-parameters. The default settings are recommended because it consistently provides good trade-offs and low query latencies. For CostOpt, preprocessing factor  $c_0$  represents the constant overhead of tree traversals and data structure settings. When migrating to a new environment, it requires only a one-time calibration, which is similar to other cost constants in optimizer. Partition granularity  $d$  directly influences the  $O(d^3)$  dynamic programming algorithm. We find that while increasing  $d$  can do produce slightly better estimation, it makes optimization too expensive with limited return, which leads to increasing in query latency. So, we recommend using our default settings for  $d$ . However, if a user finds that a particular  $d$  does not work too well, then they may want to maximize  $d$  until optimization overhead becomes unacceptable (as projected based on the  $O(d^3)$  complexity). In contrast, Greedy are more sensitive to its hyper-parameters because the decrease of estimator variance in each stratification step highly depends on the data distribution. In real deployment, we recommend using a reference

workload or historical workload to identify the hyper-parameter that works best overall.

## References

- [1] Patrick. Billingsley. 2012. *Probability and measure* (anniversary edition. ed.). Wiley, Hoboken, N.J.